

Intro to Machine Learning MCQ Quiz 1**Term:** Fall 2025

Date: September 8, 2025

Name: _____

Student ID: _____

Instructions

- Answer **all** questions. Select **exactly one** option per question.
- No books, notes, or electronic devices. Show work only where requested.
- Mark answers clearly. Illegible or multiple marks will be treated as incorrect.
- Duration: 10 minutes.

- In a data table (or design matrix), what does each *column* typically represent?
 - An example
 - A feature
 - A hypothesis
 - A loss value
 - A parameter
- How is a bias indirectly included in the feature vector?
 - $x = (x_1)$
 - $x = (0, x_1)$
 - $x = (1, x_1)$
 - $x = (x_0, x_1, x_2)$
 - $x = (1)$
- Choose the most general linear hypothesis with bias for one feature x :
 - $h_\theta(x) = \theta_1 x$
 - $h_\theta(x) = x^2 + \theta_1$
 - $h_\theta(x) = \theta_0 + \theta_1 x$
 - $h_\theta(x) = \sin x + \theta_0$
 - $h_\theta(x) = \theta_0 \theta_1 x$
- The inner product $\theta^\top x$, has what shape when $\theta, x \in \mathbb{R}^{d+1}$?
 - $(d+1) \times (d+1)$ matrix
 - $(d+1)$ -vector
 - Scalar
 - n -vector
 - Upper-triangular matrix
- The residual for example i is defined as:
 - $y_i + h_\theta(x_i)$
 - $y_i / h_\theta(x_i)$
 - $x_i^\top \theta$
 - $h_\theta(x_i) - y_i$
 - $|h_\theta(x_i) - y_i|$
- Why not minimize $\sum_i (h_\theta(x_i) - y_i)$ directly (no absolute/square)?
 - It is non-convex
 - Terms can cancel out, hiding poor fit
 - It is undefined for negative values
 - It always diverges
 - It equals zero only at the minimum
- For a *single* example (x, y) , hypothesis $h_\theta(x) = \theta^\top x$, and squared loss, $\nabla_{\theta} \frac{1}{2} (h_\theta(x) - y)^2$ equals:
 - $(h_\theta(x) - y) x$
 - $(h_\theta(x) - y)$
 - x
 - y, x
 - θ
- Batch Gradient Descent update for vector θ is:
 - $\theta \leftarrow \theta + \alpha \nabla_{\theta} J(\theta)$
 - $\theta \leftarrow \theta - \alpha \nabla_{\theta} J(\theta)$
 - $\theta \leftarrow \alpha, \theta$
 - $\theta \leftarrow \theta - J(\theta)$
 - $\theta \leftarrow \theta + \alpha J(\theta)$
- Geometry: for two fitting parameters (θ_0, θ_1) with squared loss, J is shaped like a:
 - Plane
 - Cone
 - Paraboloid
 - Hyperboloid
 - Cylinder
- Compare batch gradient descent vs. stochastic gradient descent (qualitatively):
 - Batch: slow steps; SGD: noisy but cheaper steps
 - Batch: random walk; SGD: straight line
 - Both identical in path
 - SGD always diverges
 - Batch uses one sample, SGD uses all

I certify that I have neither given nor received unauthorized assistance.

Signature: _____